

Ana Lorena Ortiz Loyola

Frankfurt am Main, Germany

☎ +49 176 62145797 • ✉ a01750283@tec.mx • [LinkedIn](#)

Data science student (Tecnológico de Monterrey, expected 2027) with hands-on experience implementing and validating predictive models across healthcare, environmental, and cryptography domains using Python, Scikit-Learn, Keras, and TensorFlow. Combines this with practical financial-services data engineering experience at Deutsche Bundesbank (AnaCredit) and daily use of AI-assisted development tools (Claude Code) to accelerate model and pipeline development.

Core Competencies

- **Big Data & Advanced Analytics:** Large-scale data processing, statistical anomaly detection, predictive modeling
- **Machine Learning:** Scikit-Learn, Keras, TensorFlow, Random Forest, SVM, K-Means, classification, clustering
- **Programming & Data:** Python (Pandas, NumPy), SQL, R, Java, C++
- **Financial Services Experience:** Data quality automation and regulatory reporting (Deutsche Bundesbank AnaCredit)
- **AI-Assisted Development:** Daily use of AI coding tools (Claude Code) for pipeline and model development
- **Cloud & Tools:** Azure Cognitive Services, AWS, Git

Professional Experience

- **Deutsche Bundesbank (AnaCredit)** **Frankfurt am Main, Germany**
Data Engineer / Intern *04/2026–07/2026*
 - Developed automated migration tests (row counts, referential integrity, aggregate consistency) using Pandas DataFrames to validate business logic after data migration
 - Built automated error detection identifying statistical deviations (duplicate IDs, value-range violations) in large-scale reported banking data

Projects

- **Tecnológico de Monterrey**
Predictive Modeling – Healthcare Optimization 10/2024–12/2024
 - Analyzed 25+ years of national health data to identify patient risk profiles and resource bottlenecks
 - Built and validated predictive models (Random Forest, Neural Networks) to forecast critical outcomes; applied K-Means clustering to segment high-risk groups
- **Tecnológico de Monterrey**
Air Quality Analysis Mexico City – SVM Classification 04/2023–06/2023
 - Built SVM models (linear and RBF kernels) to classify critical ozone levels from citywide NO_x/O₃ data with high accuracy
- **Tecnológico de Monterrey**
Lattice-Based Cryptography Research 10/2024–02/2025
 - Implemented and analyzed the complexity of lattice-based cryptographic algorithms (LWE, NTRU) in Python; contributed to a scientific publication
- **Personal Project (GitHub), built with Claude Code**
Banking Data Pipeline – Data Cleaning & Audit Log 05/2026
 - Built an end-to-end pipeline for simulated bank transactions (150 accounts) with a fully auditable data-cleaning log

Education

- **Tecnológico de Monterrey** **Mexico**
BSc in Data Science 2022–2027
Grade: 1.3 (Academic Excellence Scholarship; DAAD KOSPIE Scholarship). Topics: machine learning & AI, data analysis & statistics, numerical optimization.
- **Universität Göttingen** **Germany**
Exchange Semester 2025–2026

Languages

- German (C1), English (C1, TOEFL iBT 105/120), Spanish (native).

Publications

- Contributed to a scientific publication on lattice-based cryptography (LWE, NTRU security parameter and runtime analysis), with Prof. A. F. De Abiega L'Eglise, Tecnológico de Monterrey (citation pending publication).

Honors and Awards

- **Academic Excellence Scholarship** – Tecnológico de Monterrey.
- **DAAD KOSPIE Scholarship**.
- **National Physics Olympiad** – Mexico (2020).
- **Honorable Mention, Physics Olympiad** (2021).

References

- Available upon request.